

ADDENDUM TO VERIFICATION AND COMMENTARY: THE BAEZ-EGAN EMBEDDING OF THE LEECH LATTICE IN THE EXCEPTIONAL JORDAN ALGEBRA THE $L/2L$ PARALLEL AND HOW OUR WORK BUILDS ON IT

CLAUDE OPUS 4.7

ABSTRACT. This is a dated addendum to the original 16 April 2026 Verification and Commentary on the Baez–Egan embedding ([paper/Baez-Egan_Leech_Verification.pdf](#), 8 pp). Re-reading that work in light of our subsequent paper *An order on the Leech lattice from a $\mathbb{Z}_3$ -symmetric triple-octonion product* (v4, 24 May 2026), we notice an explicit lattice-algebraic parallel between Baez/Egan’s “key lemma” (Part 9, $L_1 \cap L_2 = 2L_0$) and Wilson’s identity ($L\bar{s} \cap Ls = 2L$), and we identify exactly how Section A.1 of our paper (*the mod-2-quotient reading*) builds on top of that shared identity. The purpose of this addendum is to credit the shared lattice-algebraic foundation correctly and to delineate what is new in our use of it.

1. THE SHARED KEY LEMMA

Both Wilson and Baez/Egan rest on the same lattice-algebraic identity, in different notations:

$$\begin{array}{ll} \text{Baez/Egan (Part 9, Section 4 of original PDF):} & L_1 \cap L_2 = 2L_0, \\ \text{Wilson (2009, Section 2):} & L\bar{s} \cap Ls = 2L. \end{array}$$

In Baez/Egan’s setup, $L_0 \cong E_8$ is self-dual; $L_1, L_2 \cong \sqrt{2}E_8$ are sublattices of L_0 arising from a common rotation R , and the duality argument $\text{dual}(L_1 \cap L_2) = \frac{1}{2}L_1 + \frac{1}{2}L_2 = \frac{1}{2}L_0$ gives $L_1 \cap L_2 = 2L_0$ (Egan).

In Wilson’s setup, $L = D_8^+$ is the octonion-realised E_8 ; the sublattices $L\bar{s}$ and Ls are the images of left-multiplication by $\bar{s}$ and s respectively (so each is isomorphic to $\sqrt{2}E_8$, since $N(s) = N(\bar{s}) = 2$), and Wilson states the identity directly.

The two identities are the same algebraic fact about E_8 and a pair of $\sqrt{2}E_8$ sublattices: Wilson’s version is the octonion-specific instance; Baez/Egan’s is the geometric stripping of it (Egan’s Part 9). Either form was already in the literature before this project began.

2. WHAT BAEZ/EGAN USE IT FOR, AND WHAT THEY DID NOT DO

Baez/Egan use the key lemma in Part 9 as a step in establishing that the lattice

$$L_L = \{(a, b, c) \in L_0^3 : a, b, c \in L_0; a+b, a+c, b+c \in L_1; a+b+c \in L_2\}$$

is isometric to $\sqrt{2}\Lambda$. The identity $L_1 \cap L_2 = 2L_0$ is used in showing both (i) L_L is even and self-dual, and (ii) L_L has no vector of squared norm < 4 . This is the use in Egan’s Part 9 of the original n -Category Café series.

Baez/Egan do *not*, anywhere in the Part 9 / Part 10 sequence, descend the octonion product on $L_0 = E_8$ modulo $2L_0$ to read $L_0/2L_0$ as an $\mathbb{F}_2$ -bilinear algebra and analyse the sublattice images $L_1/2L_0, L_2/2L_0$ in that algebra. The key lemma is used at the level of lattices in $\mathbb{O}^3$, not at the level of an $\mathbb{F}_2$ -algebra structure inside $L_0/2L_0$.

3. WHAT OUR SECTION A.1 BUILDS ON THE SAME IDENTITY

Section A.1 of our paper (*The mod-2-quotient reading of Tables A.2 and A.3*, v4) starts from the same identity, in Wilson’s form:

$$(\dagger) \quad L\bar{s} \cap Ls = 2L,$$

and uses it as the foundation for an explicitly algebraic reading. The steps:

- (1) From $(\dagger)$, $2L \subseteq L\bar{s}$ and $2L \subseteq Ls$; applying σ (which fixes $2L$, since it is a coordinate permutation of an even integer lattice), $2L \subseteq \sigma(L\bar{s})$ and $2L \subseteq \sigma(Ls)$.
- (2) $L \cdot L \subseteq L$ (Coxeter; classical) and $\mathbb{Z}$ -bilinearity descend the octonion product to an $\mathbb{F}_2$ -bilinear product $\bar{\mu}$ on $\bar{L} := L/2L \cong \mathbb{F}_2^8$.
- (3) The sublattices $\sigma(L\bar{s})$ and $\sigma(Ls)$ are full preimages of 4-dimensional $\mathbb{F}_2$ -subspaces $V := \sigma(L\bar{s})/2L$ and $W := \sigma(Ls)/2L$, and from $\sigma(L\bar{s}) + \sigma(Ls) = L$ together with $(\dagger)$ (applied after σ), $\bar{L} = V \oplus W$.
- (4) The closure statements of Lemmas 4.3 and 4.4 in our paper become, mod $2L$: V is a (two-sided) ideal of $\bar{L}$; W is a subalgebra of $\bar{L}$.
- (5) Under the natural plus-type quadratic form $q(v + 2L) := N(v)/2 \pmod{2}$ on $\bar{L}$, direct verification shows that both V and W are totally isotropic, hence Lagrangians of $(\bar{L}, q)$; the decomposition $\bar{L} = V \oplus W$ is a Witt decomposition into a complementary pair of Lagrangians.

Step (1) uses identity $(\dagger)$ directly. Steps (2)–(5) are new in our paper: they descend the octonion product to an $\mathbb{F}_2$ -bilinear product on $L/2L$, read the sublattices as ideal/subalgebra in that descended algebra, and recognise the whole picture as a polarization of an $\mathbb{F}_2$ -orthogonal space. Baez/Egan in Parts 9–10 do not take this $\mathbb{F}_2$ -algebra step.

4. ATTRIBUTION AND OVERLAP

What is shared with Baez/Egan:

- The lattice-algebraic identity $L_1 \cap L_2 = 2L_0$ (Egan’s Part 9 key lemma), equivalent to Wilson’s $L\bar{s} \cap Ls = 2L$. Egan’s duality-based proof of this is in the original PDF §4 of the 16 April 2026 Verification and Commentary, and is reproduced here for the reader’s convenience.
- The role of this identity in setting up a Leech-lattice construction inside L_0^3 via three sum-of-components conditions (Wilson 2009 in octonion language; Egan Part 9 in lattice language).

What is not in Baez/Egan and is new in our paper:

- The descent of the octonion product modulo $2L$ to an $\mathbb{F}_2$ -bilinear product on $L/2L$.
- The identification of V and W as ideal and subalgebra of this descended algebra.
- The Witt-decomposition (polarization) reading of $L/2L$ into complementary Lagrangians under the natural plus-type quadratic form.

The main paper (v4, Section A.1) cites Wilson 2009 for the sublattice identities and notes Baez/Egan in §6 as part of the related-work discussion. The cross-credit recorded here – that our Section A.1 builds on the same lattice-algebraic key lemma that Egan’s Part 9 stripped to its geometric core – is implicit in the main paper’s references but is made explicit in this addendum.

5. WHAT IT WOULD TAKE TO CLOSE THE GAP FURTHER

A natural follow-up question: does Baez/Egan’s framing extend to the $\mathbb{F}_2$ -algebra picture of Section A.1? In their setup the generalised lattice $L_L \subset L_0^3$ is constructed without explicit octonion multiplications; once those multiplications are brought back in (as in Wilson 2009, and as required for our $\bar{\mu}$ descent), the $\mathbb{F}_2$ -algebra picture is available. Egan’s Part 9 chooses to abstract away from octonion multiplication; that abstraction is what makes our $\mathbb{F}_2$ -algebra reading absent from

his account. In this sense, the present work is a continuation of the Wilson-side of the Wilson–Egan parallel, not the Egan-side.

Conversely, the Part 10 construction (lifting Λ to the 27-dimensional Jordan-algebra ambient $\tilde{\Lambda} \subset \mathfrak{h}_3(\mathbb{O})$, closed under $2(XY + YX)$) is a different direction: it embeds Λ into a 27-dimensional ring, whereas our mod-2 picture works within 24-dimensional $L/2L$ inherited from L alone. The two routes are not in competition; they are different uses of the same shared key lemma.

REFERENCES

- Original Verification PDF: `paper/Baez-Egan-Leech_Verification.pdf`, 16 April 2026 (Claude, Anthropic).
- Our main paper: *An order on the Leech lattice from a $\mathbb{Z}_3$ -symmetric triple-octonion product*, v4 (24 May 2026), Section A.1 “The mod-2-quotient reading of Tables A.2 and A.3”.
- Wilson, R. A., *Octonions and the Leech lattice*, J. Algebra **322** (2009), 2186–2190.
- Baez, J. C. (with G. Egan and others), *Integral octonions* (parts 9 and 10), *n-Category Café*, November–December 2014.

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